strate that current systems are not ready for AI-only research-the integrity of science depends on maintaining rigorous human evaluation as AI capabilities advance.

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A Supplementary

A.1 Stratified Sampling Procedure

We implement the stratified sampling pipeline to construct the calibration corpus as follows.

First, we partition the score space using bin edges $t_0 < \cdots < t_B$ to define score bins $B_b = [t_{b-1}, t_b)$ for $b = 1, \ldots, B$.

For each bin–status combination $(b, c) \in \{1, \ldots, B\} \times \mathcal{C}_{\text{stat}}$, we define:

$$\begin{aligned} \mathcal{I}_{b,c} &= \{i : h_i \in B_b, \sigma_i = c\}, \\ N_{b,c} &= |\mathcal{I}_{b,c}|, \quad p_{b,c} = \frac{N_{b,c}}{N_\star}, \end{aligned} \tag{5}$$

where $N_\star = \sum_{b=1}^{B} \sum_{c \in \mathcal{C}_{\text{stat}}} N_{b,c}$ is the total reference pool size.

Given a target calibration size N_{cal}, we allocate samples to each cell using proportional allocation with the largest-remainder method:

$$\begin{aligned} n'_{b,c} &= p_{b,c} N_{\text{cal}}, \quad n_{b,c} = \lfloor n'_{b,c} \rfloor \\ R &= N_{\text{cal}} - \sum_{b,c} n_{b,c}. \end{aligned}$$

We then add one additional sample to the R cells with the largest remainders $n'_{b,c} - \lfloor n'_{b,c} \rfloor$.

Finally, we sample uniformly without replacement $\mathcal{S}_{b,c} \subseteq \mathcal{I}_{b,c}$ with $|\mathcal{S}_{b,c}| = n_{b,c}$ and construct:

$$\begin{aligned} \mathcal{D}_{\text{cal}} &= \{(x_i, y_i^{\text{hum}}, \sigma_i, h_i) : i \in \mathcal{S}\}, \\ \text{where} \quad \mathcal{S} &= \bigcup_{b=1}^{B} \bigcup_{c \in \mathcal{C}_{\text{stat}}} \mathcal{S}_{b,c}. \end{aligned} \tag{6}$$

This construction ensures that $\hat{p}^{\text{cal}}_{b,c} = n_{b,c}/N_{\text{cal}} \approx p_{b,c}$ for all (b, c), preserving both score-bin and status marginals up to integer rounding.

A.2 Error Analysis of Review Scoring

Having defined our review aggregation mechanism, we now turn to a fundamental question: how reliable are the resulting scores and decisions? When

we combine judgments from multiple reviewer agents, two sources of uncertainty arise. First, each reviewer introduces randomness—even when evaluating the same paper, a model may produce slightly different scores across runs. Second, our decision thresholds are estimated from finite calibration data and therefore subject to sampling error.

We address these concerns by providing a rigorous error analysis that answers two questions:

- **Q1: How much does ensembling reduce randomness?** Under independent reviewers, we give concentration bounds in Theorem 1 and Corollary 2 to show how tightly $s(x)$ clusters around its latent mean.
- **Q2: How reliable is a threshold picked from finite calibration data?** We give bounds on the acceptance-rate estimation error and the 0.5-probability threshold with isotonic calibration in Propositions 1 and 2.

We also provide a Bayesian view that yields credible intervals for decision-making under uncertainty.

Assumptions. To make our analysis tractable, we impose two standard regularity conditions on the review process. For each model $m \in \mathcal{M}$, let $\mathbf{r}_m(x) \in \mathbb{R}^K$ denote the rubric vector and define the weighted consensus rubric $\bar{\mathbf{r}}(x) = \sum_m w_m \, \mathbf{r}_m(x)$. Let the latent mean be $\bar{\boldsymbol{\mu}}(x) = \sum_m w_m \, \mathbb{E}[\mathbf{r}_m(x) \mid x]$. We assume:

- (**Sub-Gaussian**) For each m, the centered rubric $\mathbf{z}_m(x) := \mathbf{r}_m(x) - \mathbb{E}[\mathbf{r}_m(x) \mid x]$ is *vector sub-Gaussian*: for all $\mathbf{u} \in \mathbb{R}^K$, $\langle \mathbf{u}, \mathbf{z}_m(x) \rangle$ is sub-Gaussian with proxy $\sqrt{\mathbf{u}^\top \Sigma_m \mathbf{u}}$. Moreover, $\{\mathbf{z}_m(x)\}_{m \in \mathcal{M}}$ are mutually independent.
- (**Lipschitz**) $\phi : \mathbb{R}^K \to \mathbb{R}$ is L_ϕ-Lipschitz w.r.t. ℓ_2: $|\phi(\mathbf{a}) - \phi(\mathbf{b})| \le L_\phi \|\mathbf{a} - \mathbf{b}\|_2$.

These assumptions are natural in the peer-review setting. The sub-Gaussian property follows from the fact that venues always require bounded rubric scores, ensuring $r_{m,k} \in [a_k, b_k]$ and thus sub-Gaussianity via Hoeffding's lemma (Hoeffding, 1963). The independence assumption reflects the standard practice that different reviewers evaluate papers independently without coordination. The Lipschitz condition is satisfied by common aggregation functions such as weighted averages ($\phi(\mathbf{a}) = \mathbf{v}^\top \mathbf{a}$, $L_\phi = \|\mathbf{v}\|_2$) or selecting a single overall score ($L_\phi = 1$).

With these assumptions in place, we define the latent target score $\mu_s(x) := \phi(\bar{\boldsymbol{\mu}}(x))$ as the score we would obtain if each reviewer's noise were averaged out. Under independence across reviewers, the aggregate vector noise has proxy matrix

$$\Sigma_{\mathrm{vec}}(\mathbf{w}) \;:=\; \sum_{m \in \mathcal{M}} w_m^2 \, \Sigma_m \;\in \mathbb{R}^{K \times K},$$

and we use the scalar variance proxy

$$V_w \;:=\; \lambda_{\max}\big(\Sigma_{\mathrm{vec}}(\mathbf{w})\big).$$

Frequentist concentration for ensemble scoring. We begin by quantifying how closely the observed ensemble score $s(x)$ tracks the latent mean $\mu_s(x)$. The following result shows that aggregating multiple independent reviewers yields exponentially tight concentration.

Theorem 1 (Bernstein-McDiarmid concentration and margin bound). Under the assumptions above, let $c_m := L_\phi \, w_m \sqrt{\sum_{k=1}^K (b_k - a_k)^2}$ and $\sigma_w^2 := \mathrm{Var}[s(x)] \le L_\phi^2 \sum_m w_m^2 \, \lambda_{\max}(\Sigma_m)$, with $c_{\max} := \max_m c_m$. Then for any $t > 0$,

$$\Pr\big(s(x) - \mu_s(x) \ge t\big) \;\le\; \exp\Big(-\frac{t^2}{2\sigma_w^2 + \frac{2}{3} c_{\max} t}\Big). \tag{7}$$

Consequently, with $y^\star(x) = \mathbb{I}[\mu_s(x) \ge \tau]$ denoting the latent decision at threshold τ and $\gamma(x) = |\mu_s(x) - \tau|$ denoting the margin,

$$\Pr\big(\hat{y}(x) \ne y^\star(x)\big) \;\le\; \exp\Big(-\frac{\gamma(x)^2}{2\sigma_w^2 + \frac{2}{3} c_{\max} \gamma(x)}\Big). \tag{8}$$

Corollary 1 (Variance-minimizing weights for linear aggregation). Suppose $\phi(\mathbf{a}) = \mathbf{v}^\top \mathbf{a}$ is linear. Let $c_m := \mathbf{v}^\top \Sigma_m \mathbf{v}$. Then $V_w = \sum_m w_m^2 c_m$ and among $\mathbf{w} \in \Delta^{M-1}$ the bound in (8) is minimized by

$$w_m^\star \;\propto\; \frac{1}{c_m} \;=\; \frac{1}{\mathbf{v}^\top \Sigma_m \mathbf{v}},$$

i.e., (diagonal) GLS/precision weighting in the projected variance.

Scalar-score simplification (overall assessment). The general vector-rubric framework of Theorem 1 applies when reviewers provide detailed multi-criterion scores. However, in many venues (e.g., ICLR/ICML), reviewers independently provide a single bounded *overall assessment* that already aggregates rubric criteria internally. This special case